

Chain Rule 02: Logarithmic of Sine Composite Function

Course: Mathematics-I (Differential Calculus)

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প্রদত্ত সমীকরণ: $y = \log \sin x$ নির্ণেয়: $\frac{dy}{dx}$

ENGLISH NOTE — LOGARITHMIC-TRIGONOMETRIC CHAIN RULE FORMULATION

The problem presents a composite function where the argument of the logarithm is the sine function:

$$f(u) = \log u (\equiv \ln u), \quad u = g(x) = \sin x$$

Applying Leibniz's Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{\sin x} \cdot \frac{d}{dx}(\sin x) = \frac{1}{\sin x} \cdot \cos x = \cot x$$

১. গাণিতিক কাঠামো ও প্রয়োজনীয় সূত্রসমূহ

উপাদান (Component)	ফাংশন রূপ	প্রমিত সূত্র (Standard Formula)
বহিঃস্থ ফাংশন (Outer)	$\log u$	$\frac{d}{du}(\log u) = \frac{1}{u}$
অভ্যন্তরীণ ফাংশন (Inner)	$u = \sin x$	$\frac{du}{dx} = \frac{d}{dx}(\sin x) = \cos x$
ত্রিকোণমিতিক অনুপাত	$\frac{\cos x}{\sin x}$	$\frac{\cos x}{\sin x} = \cot x$

ENGLISH NOTE — CRITICAL AMIE EXAM TRAPS & PITFALLS

1. **Omitting Inner Derivative:** A frequent mistake is writing only $\frac{1}{\sin x}$ without completing the Chain Rule with $\frac{d}{dx}(\sin x) = \cos x$.

2. **Standard Ratio Form:** Always simplify $\frac{\cos x}{\sin x}$ into $\cot x$. Leaving it unreduced may result in minor mark deductions in AMIE Section-A.

২. ধাপে ধাপে পূর্ণাঙ্গ সমাধান (Step-by-Step Solution)

ধাপ ১: উভয় পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করে:

$$\frac{dy}{dx} = \frac{d}{dx}(\log \sin x)$$

ধাপ ২: শৃঙ্খল নিয়ম (Chain Rule) অনুসারে সাজানো:

$$\frac{dy}{dx} = \frac{1}{\sin x} \cdot \frac{d}{dx}(\sin x)$$

ধাপ ৩: অভ্যন্তরীণ পদ $\sin x$ -এর অন্তরজ $\cos x$ বসিয়ে:

$$\frac{dy}{dx} = \frac{1}{\sin x} \cdot \cos x = \frac{\cos x}{\sin x}$$

ধাপ ৪: ত্রিকোণমিতিক অনুপাত কোট্যানজেন্টে (cot x) রূপান্তর:

$$\frac{dy}{dx} = \cot x$$

চূড়ান্ত উত্তর (FINAL STANDARD ANSWER)

$$\frac{dy}{dx} = \cot x$$

🔬 ENGLISH NOTE — DOMAIN, SINGULARITIES & INTEGRAL DUALITY

- Domain of $y(x)$:** Logarithmic argument must be strictly positive in real calculus: $\sin x > 0$. Thus, $x \in (2k\pi, (2k+1)\pi)$, $k \in \mathbb{Z}$ (1st and 2nd quadrants):
$$\text{Dom}(y) = \bigcup_{k \in \mathbb{Z}} (2k\pi, (2k+1)\pi)$$
- Fundamental Integral Duality:** This fundamental differential proves the standard calculus integral:
$$\int \cot x \, dx = \log |\sin x| + C$$

💡 ENGLISH NOTE — FORMAL SUBSTITUTION MODEL

Let $u = \sin x \implies \frac{du}{dx} = \cos x$.

$$y = \log u \implies \frac{dy}{du} = \frac{1}{u}$$
$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{u} \cdot \cos x = \frac{\cos x}{\sin x} = \cot x$$

This 2-step substitution is rigorous and textbook-standard for AMIE examinations.

🎯 AMIE EXAM MARKS DISTRIBUTION STRATEGY

Exam Evaluation Step	Required Mathematical Rigor	Marks
Chain Rule Formulation	Writing $\frac{1}{\sin x} \cdot \frac{d}{dx}(\sin x)$	2.0
Sine Differentiation	Accurate evaluation of $\frac{d}{dx}(\sin x) = \cos x$	1.5
Cotangent Simplification	Writing final canonical form $\cot x$	1.5
Total AMIE Section-A Marks		5.0 / 5.0